Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Geometrical interpretation of Partial derivatives

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Geometrical interpretation of Partial derivatives

1 Geometrical interpretation of Partial derivatives

Study focus: Geometrical interpretation of Partial derivatives. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We were discussing about the differentiability of a multivariable function and we call that as a partial derivatives.

Key details

- We also defined what is a higher order derivatives and what is the mixed order partial derivative.
- How to find ZX, how to find ZY, and if you had observed we actually did these many second order partial derivatives, right?
- But of course these were equal, but still.
- second order derivative, which are Zx, ZYY, ZXY, and ZYX.
- in M to the power of N order.
- That means you the first order derivative will have two terms.
- The second order will have.
- Again four terms for a two variable and so like that it continues.
- Then what is the dy by dx represents?
- that the derivative always gives me the change, right?
- But change in what?
- That's for a single variable function.
- I have a points which lie on the circle.
- You see a circle in the bottom?
- And that lie on this curve.
- It gives you the slope of the tangent.
- Something like this, and if you have a point, at this point, you please don't think that I am intersecting, this is a tangent because it's in 3D, it looks like that.

Example or application from the lecture

- A function of two independent variables, suppose you have a function of two independent variables, which is a bivariate function, it has two first order derivatives, which are zx and zy, and it has four derivative.
- That means, suppose I have a function f which is a function okay let me call z as a function of n independent variable.
- Suppose you have a function of m independent variable, then you will have m to the power of n derivatives of order n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

So because it's a surface, at any given point you will have infinitely many tangents to the surface.

Key details

- Are you following what I'm trying to say?
- Let me erase all these things.
- There are infinitely many such tangent.
- I just want to find the slope of one such tangent.
- What is the tangent?
- That tangent is the variable with respect to which you are going to differentiate.
- Then Zx will give me, so I am calculating Z x a at X0 Y0.
- This derivative at this point will give me the slope of the tangent to the curve at x0 y naught, which is parallel to x axis.
- Are you following?
- The partial derivative with respect to y gives me the slope of the tangent.
- This gives me the slope of the tangent to this surface at this point.
- And what is the tangent?
- That tangent which is parallel to the y-axis.
- So the per first order partial derivative with respect to x gives me the slope of the tangent parallel to x axis and z y gives me the slope of the tangent parallel to the y axis.
- But the general question that would arise is.
- So that means what I if I had the my coordinate system like this and you had a surface.
- And the tangents that you got now is either it was parallel to x axis or it was parallel to y axis, something like that.
- But I want something the that is different.
- What how do I get the slope of this or what will this represent?
- This is ZX and this is Z1.
- I can resolve that into the component along x axis and the component along y axis.
- And this vector also gives me the slope of the tangent and this is called the gradient.

Example or application from the lecture

- Let's say that this is the surface which represents Z is equal to F of XY.
- So geometally the first order partial derivatives, suppose let me write this as zx and this is zy.
- Let's see how does it look like.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 10:00–14:59

As that like the first order derivative in the for one variable gives you velocity.

Key details

- And then D square y by dx square gives you acceleration.
- Do what do they call for the higher order derivative?
- T cube y by dx cube or d to the power of four y by d x power four.
- When you did maxima minima.
- It's not that easy for me to plot in 2D, but I'll show you a video.
- And then if f y is greater than zero, then it is concave up in the y direction.
- Same thing with the mixed order partial derivative.
- Fxy will tell me how the partial derivative of f x is changing in y direction.
- What is the change in f y along the x direction?
- I mean it's again, it's not that easy for you to visualize.
- This is just from Ultra Am lab.
- what I have done is I have animated that so that you can visualize for different values of fxx.
- So I have taken the cases where for the first case as you can see I have taken f of xx to be sorry negative.
- So in the first case I have taken FXX to be negative and then FYY also to be negative and then I have taken fx y to be positive.
- And then I have increased the value of x.
- And you can see that it is concave up along the y axis or x axis, sorry.
- And then I have changed now fx is positive, but I have changed x fxy to be negative.
- Yeah, it's it's nice to see these visuals.
- you can try by yourself by changing different values.
- just to check what happens.
- and this negative and this positive.
- And you can keep on changing the value of X and changing the value of Y.
- That will help you to visualize the tangent.
- Do you see the tangent here?
- , so this these things will be better if you try by yourself.
- in the next class what we are going to do is we'll try and solve some problems on partial derivatives.
- We'll try to find mixed order partial derivatives, we'll try to verify the Claret theorem and all those things.

Example or application from the lecture

- The second derivative so suppose what I mean is suppose you have y is equal to f of x, then dy by dx gives you velocity.
- But you also know that suppose you want to find whether the curve is concave up or concave down, you use usually the second derivative, right?
- Suppose I have the second order partial derivative which is fxx and if this is greater than zero, that means the function is concave up in the x direction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 15:00–19:59

I want you to explore.

Key details

- I'll give you a link in the end of the class wherein you can try all these things.
- You can try how to plot a three dimensional graph, how to do this animation, and, how to check the partial derivative and his geometrical interpretation and other things.

Example or application from the lecture

- But I have given a link in my next slide.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Geometrical interpretation of Partial derivatives transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Geometrical interpretation of Partial derivatives transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice